

Household Behavior over the Life Cycle

Assignment 1: Labor Supply and Children

1. What value of β_1 produces an event-study graph of working hours with something close to a drop in hours of 10pct around the time of birth as in Kleven, Landaís and Søgaaard (2019, Fig. 1B)? Be explicit about how you found the value.

β_1 is a "child penalty" parameter, meaning that when you get a child, the stress of juggling few hours of sleep and the guilt of not being at home gives you disutility of working. The individual responds optimally by working fewer hours to reduce that extra disutility.

To find the correct β_1 value I searched over a grid of values $\beta_1 \in \{0.05, 0.055, 0.06, 0.065, 0.1, 0.5, 1.0, 1.5, 2.0\}$ and found that $\beta_1 = 0.056$ produces a drop in hours of 10% around the time of birth. This is consistent with the findings in Kleven et al. (2019, Fig. 1B), where the initial drop in hours worked for women is approximately 13%. Figure 1 shows the event-study graph of hours worked relative to the period before birth. This illustrates the drop in hours of 10 pct immediately after the child is born, and afterwards increasing again as the child grows older.

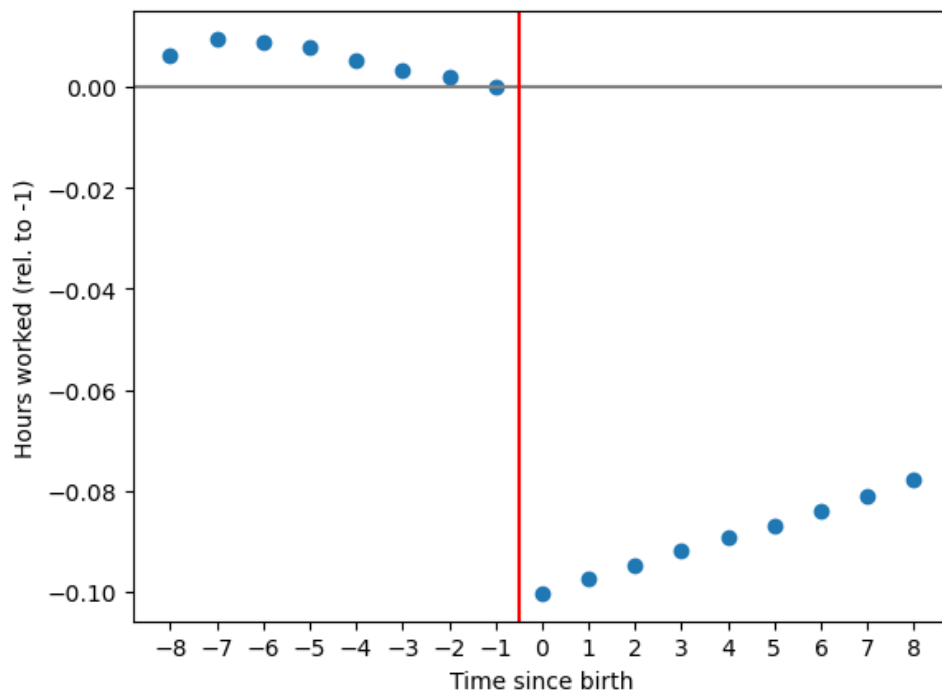


Figure 1: Hours worked

2. Introduce in the model a spouse who contributes with income y_t every period. Let the income depend on age/period, such that $y_t = 0.1 + 0.01 \cdot t$. Write out the affected equations and implement it in Python. Compare and comment on the simulated behavior from this alternative model in relation to that of the baseline model above. Concretely plot age profiles of e.g. labor supply, consumption, fertility and savings.

[$y_t = 0 \forall t$ gives the baseline model.]

Only the budget constraint is affected. Here we simply add the spouse's income y_t to a_{t+1}

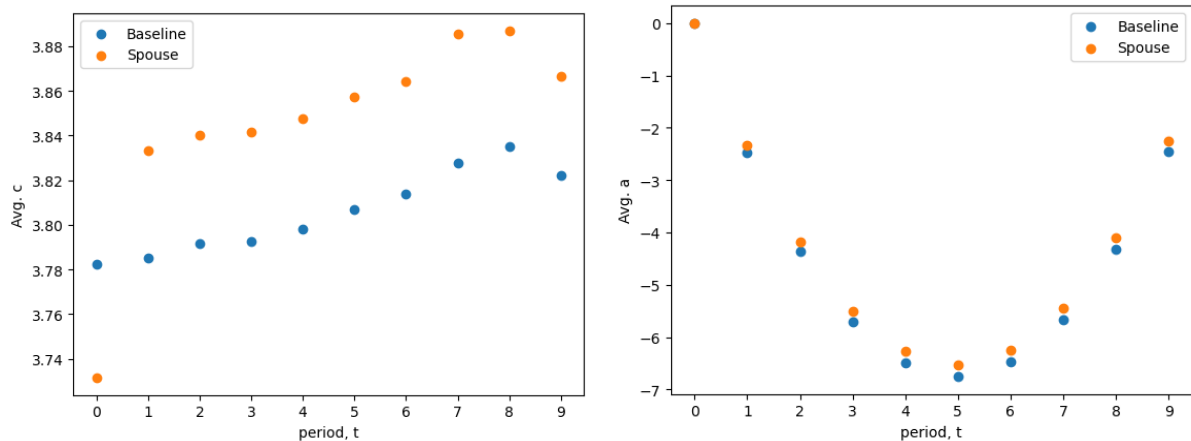
as shown below:

$$a_{t+1} = (1 + r)(a_t + (1 - \tau)w_t h_t + y_t - c_t)$$

Where $y_t = 0.1 + 0.01 \cdot t$.

Figure 2 illustrates the four figures for how consumption (c), savings (a), labor supply (h) and fertility (n) is affected when a spouse is included in the model.

- Looking at consumption we notice that with a spouse, consumption is higher in all periods compared to the baseline model, which reflects the added household income y_t . The initial jump from period 0 to period 1 could reflect consumption smoothing, i.e. the individual anticipates future spouse income and borrows against it to increase consumption already from period 1 and onwards.
- The savings graph shows negative savings for both the baseline and the spouse model which indicates that they are borrowing. The spouse model borrows relatively less than the baseline, as the household has more income in this model.
- Looking at the labor supply we see that the spouse model contributes with fewer working hours. Illustrates an income effect as the extra income from the spouse means that you don't have to work as much yourself.
- Fertility is the same with and without a spouse. Makes sense as the spouse only affects the budget constraint plus the fertility is also constant.



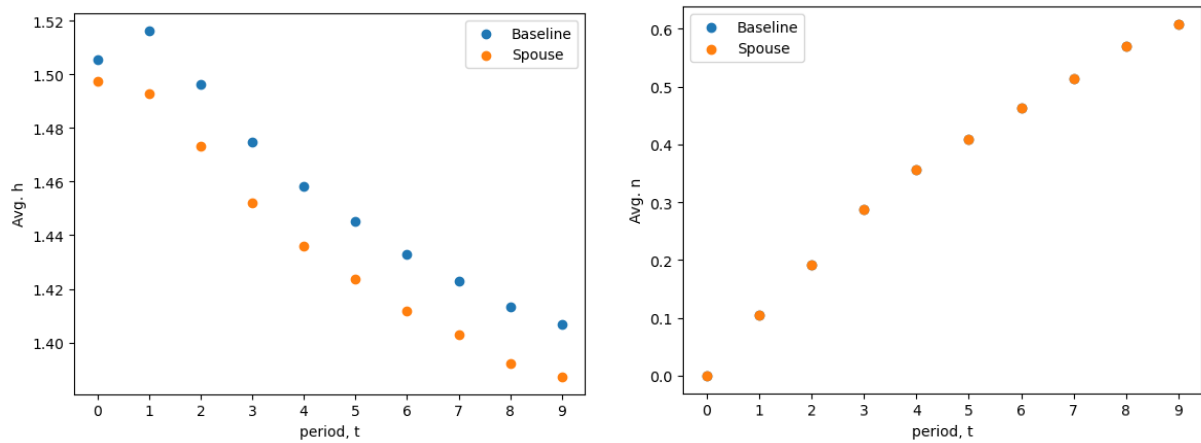


Figure 2: Baseline and spouse model

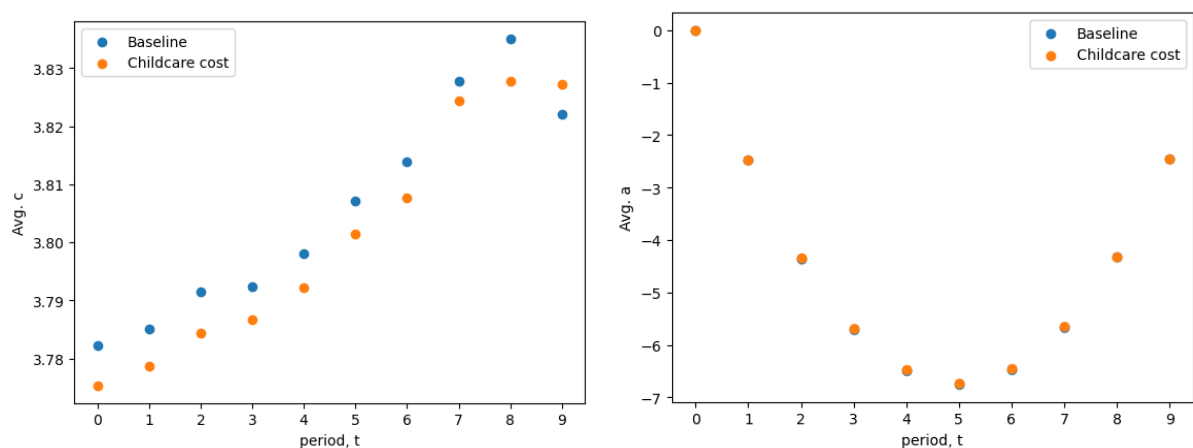
3. Introduce in the model a childcare cost of $\theta = 0.05$ in the presence of a child. You can think of this as a policy reform reducing the child care subsidy. Write out the affected equations and implement it in Python. Compare the simulated behavior from this alternative model in relation to that of the baseline model above. Compare also the Marshall elasticity in both models and comment on how the income and substitution effects differs (you do not need to calculate them).

[$y_t = 0 \forall t$ and $\theta = 0$ gives the baseline model.]

Only the budget constraint is affected. Childcare cost θ is subtracted from the income:

$$a_{t+1} = (1 + r)(a_t + (1 - \tau)w_t h_t - c_t - \theta \cdot n_t)$$

Where $y_t = 0.1 + 0.01 \cdot t$.



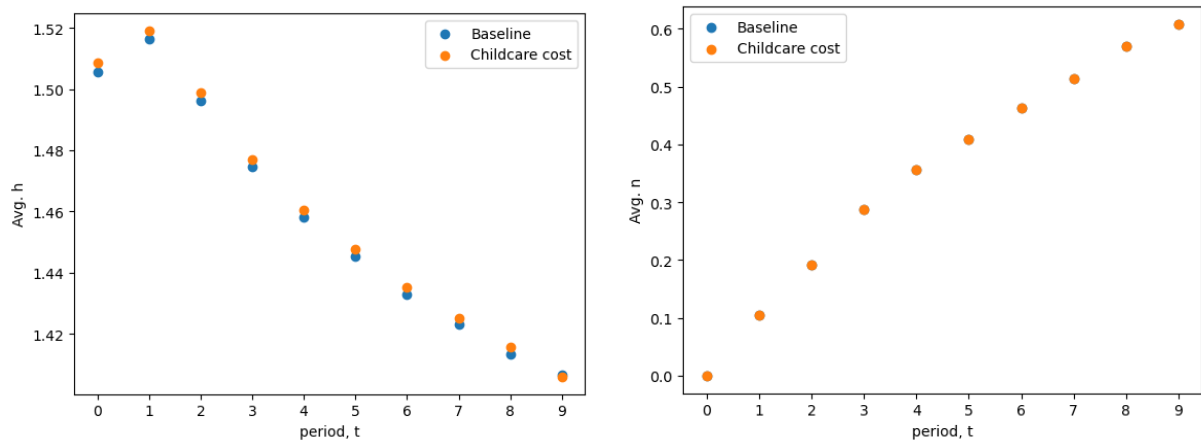


Figure 3: Baseline and childcare cost model

Figure 3 illustrates how consumption (c), savings (a), labor supply (h) and fertility (n) is affected when a child and therefore also childcare costs are included in the model.

- The childcare cost model has lower consumption in all periods. The extra cost of taking care of the child reduces the household's disposable income, so they can afford less.
 - Practically identical with the baseline. Intuitively this does not really make sense.
 - The childcare cost model works very few hours more than the baseline. Here it is the substitution effect that dominates. The individual works more to compensate for the extra cost and thus the extra income lost.
 - Fertility is still exogenous with a constant probability why introducing the childcare cost does not affect the chance of reproducing.
4. *Discuss how endogenous fertility adjustments could affect the impact of a child subsidy reform as the one above. Specifically, think about substitution and income effects related to fertility.*

If the fertility was endogenous a child subsidy reform would have two opposing effects on labor supply.

Direct effect: Reducing θ means that having a child becomes cheaper. This lowers the financial burden of having a child making more room in the household budget. This gives individuals an incentive to work more and thus supply more labor since the cost of having a child no longer weighs as heavily on the household budget.

Indirect effect: However, as children are now cheaper they are now seen as more attractive compared to other goods, which increases the demand for children and therefore also fertility. That is the substitution effect. Furthermore, since the subsidy also increases real household income, then the higher income pushes fertility even higher assuming children are a normal good. That is the income effect.

The total effect on labor supply depends on which of the above effects dominates. If the households respond by having significantly more children, the negative indirect effect could dominate the direct positive effect resulting in labor supply declining. In our model fertility is exogenous, which means that the number of children is random and not affected by θ . So when we reduce θ it is only the direct effect working - people working more because children are now relatively cheaper. But in reality people would also have more children as a response to the subsidy reform and thus work less (due to maternity leave etc). Our model doesn't capture this, meaning the increase in labor supply is too big - the model overestimates the effect of the reform.

5. *Imagine that whether an income-generating spouse is present or not is random and $p_s = 0.8$ is the likelihood that a spouse is present in a given period. Furthermore, imagine that a child can only arrive next period if a spouse is currently present. Formulate mathematically the recursive formulation of this model and implement it in Python. Compare the simulated behavior from this alternative model in relation to that of the baseline model above.*

[$y_t = 0 \forall t$, $\theta = 0$ and $p_s = 1$ gives the baseline model.]

So we now have an additional state variable, which is if a spouse is present or not $s_t \in 0, 1$ with probability $p_s = 0.8$. The Bellman equation is:

$$V_t(n_t, a_t, k_t) = \max_{c_t, h_t} \frac{c_t^{1+\eta}}{1+\eta} - \beta(n_t) \frac{h_t^{1+\gamma}}{1+\gamma} + \rho \mathbb{E}_t[V_{t+1}(n_{t+1}, a_{t+1}, k_{t+1}, s_{t+1})]$$

We now add the spouse income to the budget constraint. The spouse income y_t only enters when $s_t = 1$:

$$a_{t+1} = (1+r)(a_t + (1-\tau)w_t h_t + s_t y_t - c_t)$$

The probability of a child arriving now looks like:

$$p(n_t, s_t) = \begin{cases} p_n & \text{if } n_t = 0 \text{ and } s_t = 1 \\ 0 & \text{otherwise} \end{cases}$$

The spouse presence evolves each period:

$$s_{t+1} = \begin{cases} 1 & \text{with probability } p_s \\ 0 & \text{with probability } 1 - p_s \end{cases}$$

Calculating the expected future value we need to take into account whether a spouse is present (s_{t+1}) and if a child is arriving (n_{t+1}). We have the following possible outcome combinations:

If a spouse is present next period (with probability p_s):

- Child arriving with probability $p(n_t, s_t) \rightarrow V_{t+1}(n_t + 1, a_{t+1}, k_{t+1}, s' = 1)$
- No child arriving with probability $1 - p(n_t, s_t) \rightarrow V_{t+1}(n_t, a_{t+1}, k_{t+1}, s' = 1)$

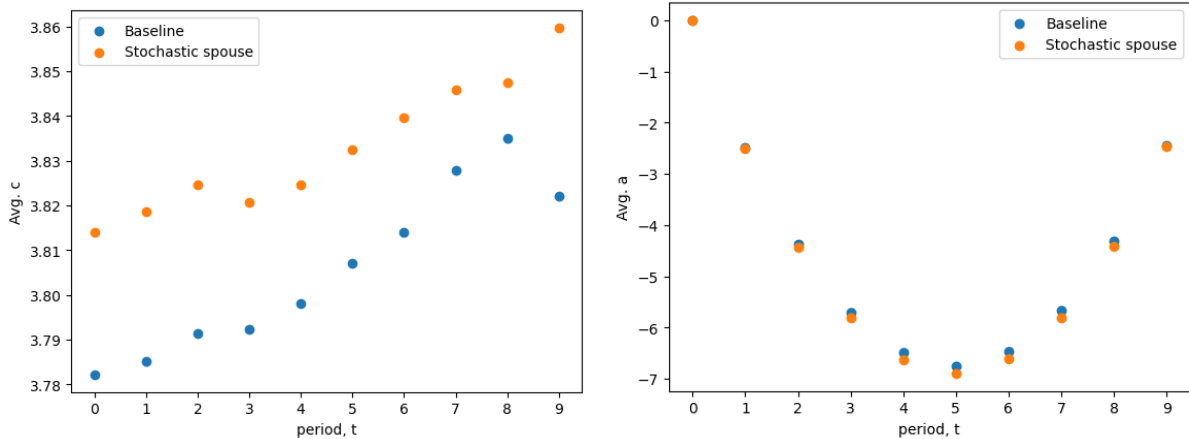
If a spouse is not present next period (probability $1 - p_s$):

- Child arriving with probability $p(n_t, s_t) \rightarrow V_{t+1}(n_t + 1, a_{t+1}, k_{t+1}, s' = 0)$
- No child arriving with probability $1 - p(n_t, s_t) \rightarrow V_{t+1}(n_t, a_{t+1}, k_{t+1}, s' = 0)$

The above combinations is then weighted with each of their probabilities which can be written as.

$$E_t[V_{t+1}] = p_s p(n_t, s_t) V_{t+1}(n_t + 1, a_{t+1}, k_{t+1}, 1) + p_s (1 - p(n_t, s_t)) V_{t+1}(n_t, a_{t+1}, k_{t+1}, 1) + (1 - p_s) p(n_t, s_t) V_{t+1}(n_t + 1, a_{t+1}, k_{t+1}, 0) + (1 - p_s) (1 - p(n_t, s_t)) V_{t+1}(n_t, a_{t+1}, k_{t+1}, 0)$$

Looking at the consumption in this stochastic spouse model we see that the consumption is higher compared to the baseline in all periods. Since a child is less likely to arrive the household faces lower expected future disutility from children and therefore consumes more. The savings are almost identical to the baseline with the baseline being slightly above the alternative model. The labor supply is slightly higher in the stochastic spouse model, meaning the individuals work slightly more compared to the baseline, especially in later periods. Since fewer children are expected over the lifetime in the alternative model, the average disutility of working is lower compared to the baseline. This means that individuals choose to spend more hours working explaining the higher labor supply in the stochastic spouse model. This model also has lower fertility compared to the baseline. This is because the children can only arrive when a spouse is present, which only happens with a probability of 80%, meaning fewer children are born over the life cycle compared to the baseline.



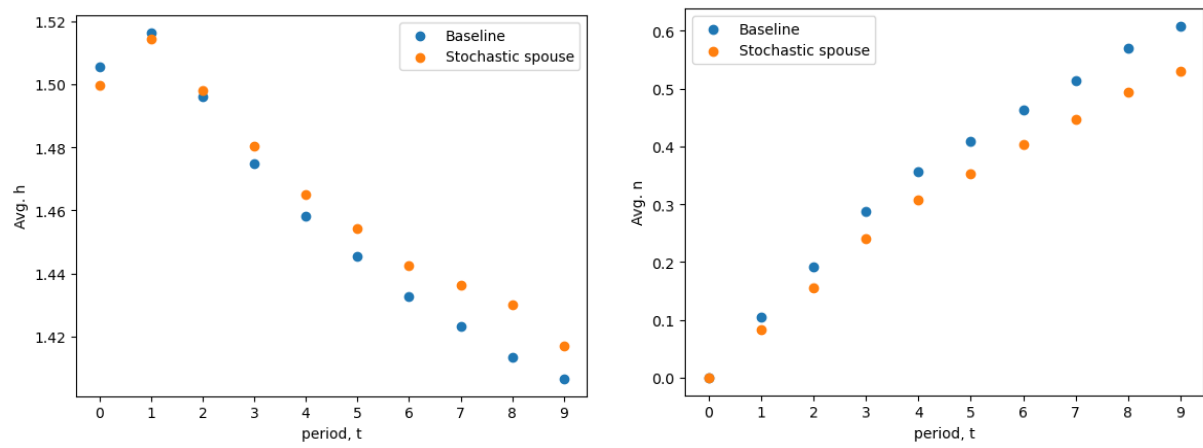


Figure 4: Baseline and stochastic spouse model